

```
# Customer Churn rate
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/Churn_Modelling (1).csv')
```

```
df.head()
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	
3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	

```
df.drop(columns= ["CustomerId", "Surname", "RowNumber"], axis= 1 , inplace= True, errors='ignore')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 11 columns):
#   Column              Non-Null Count  Dtype  
---  -
0   CreditScore          10000 non-null  int64  
1   Geography             10000 non-null  object  
2   Gender                10000 non-null  object  
3   Age                  10000 non-null  int64  
4   Tenure                10000 non-null  int64  
5   Balance               10000 non-null  float64 
6   NumOfProducts         10000 non-null  int64  
7   HasCrCard             10000 non-null  int64  
8   IsActiveMember        10000 non-null  int64  
9   EstimatedSalary       10000 non-null  float64 
10  Exited                10000 non-null  int64  
dtypes: float64(2), int64(7), object(2)
memory usage: 859.5+ KB
```

```
df.describe()
```

	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	
count	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	1
mean	650.528800	38.921800	5.012800	76485.889288	1.530200	0.70550	0.515100	100090.239881	
std	96.653299	10.487806	2.892174	62397.405202	0.581654	0.45584	0.499797	57510.492818	
min	350.000000	18.000000	0.000000	0.000000	1.000000	0.000000	0.000000	11.580000	
25%	584.000000	32.000000	3.000000	0.000000	1.000000	0.000000	0.000000	51002.110000	
50%	652.000000	37.000000	5.000000	97198.540000	1.000000	1.000000	1.000000	100193.915000	
75%	718.000000	44.000000	7.000000	127644.240000	2.000000	1.000000	1.000000	149388.247500	
max	850.000000	92.000000	10.000000	250898.090000	4.000000	1.000000	1.000000	199992.480000	

```
df.isnull().sum()
```

	0
CreditScore	0
Geography	0
Gender	0
Age	0
Tenure	0
Balance	0
NumOfProducts	0
HasCrCard	0
IsActiveMember	0
EstimatedSalary	0
Exited	0

dtype: int64

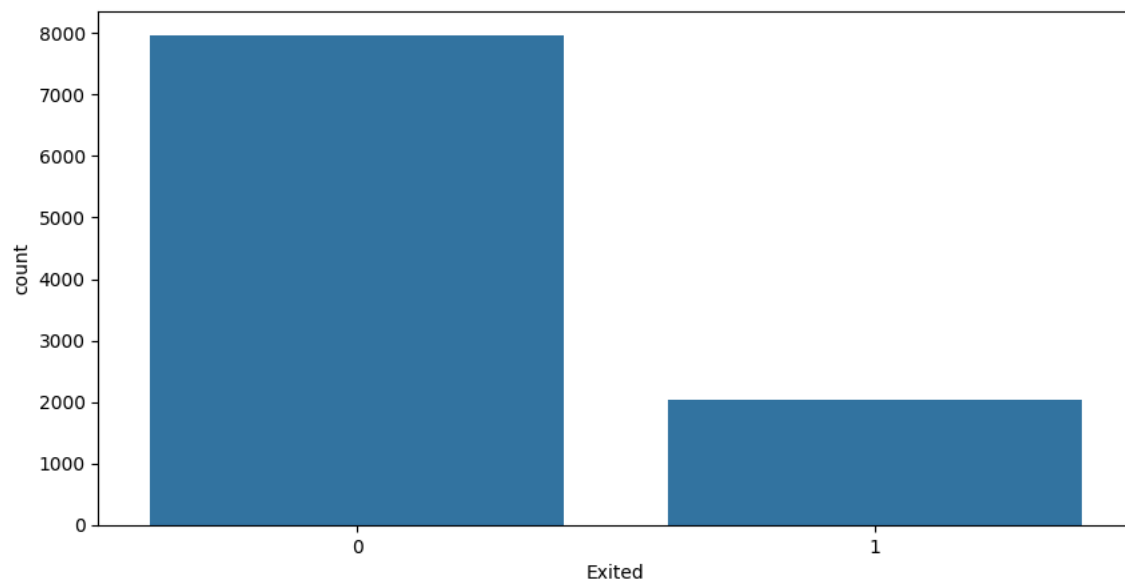
```
count = df['Exited'].value_counts()
count
```

	count
Exited	
0	7963
1	2037

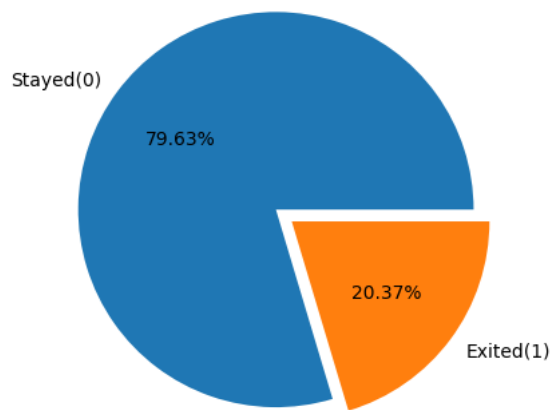
dtype: int64

Pie plot

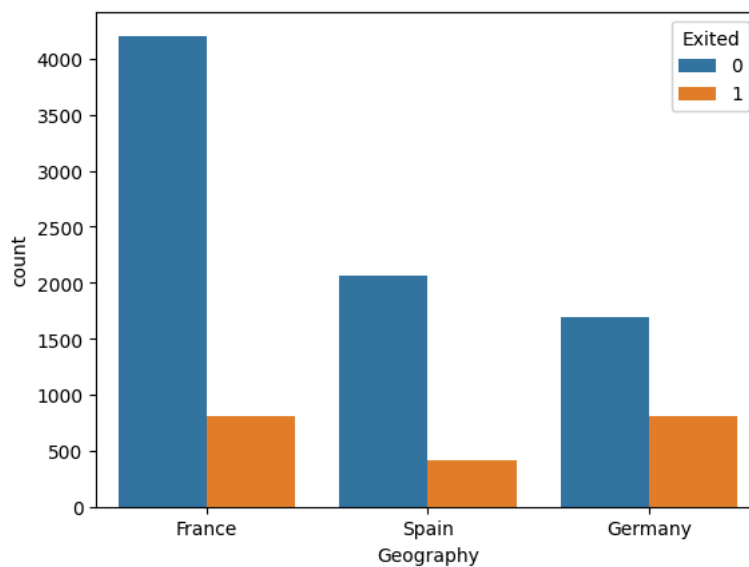
```
plt.figure(figsize=(10,5))
sns.countplot(x = 'Exited', data = df)
plt.show()
```



```
plt.pie(count, explode=(0,0.1), labels= ["Stayed(0)", "Exited(1)"], autopct= "%1.2f%%")
plt.show()
```



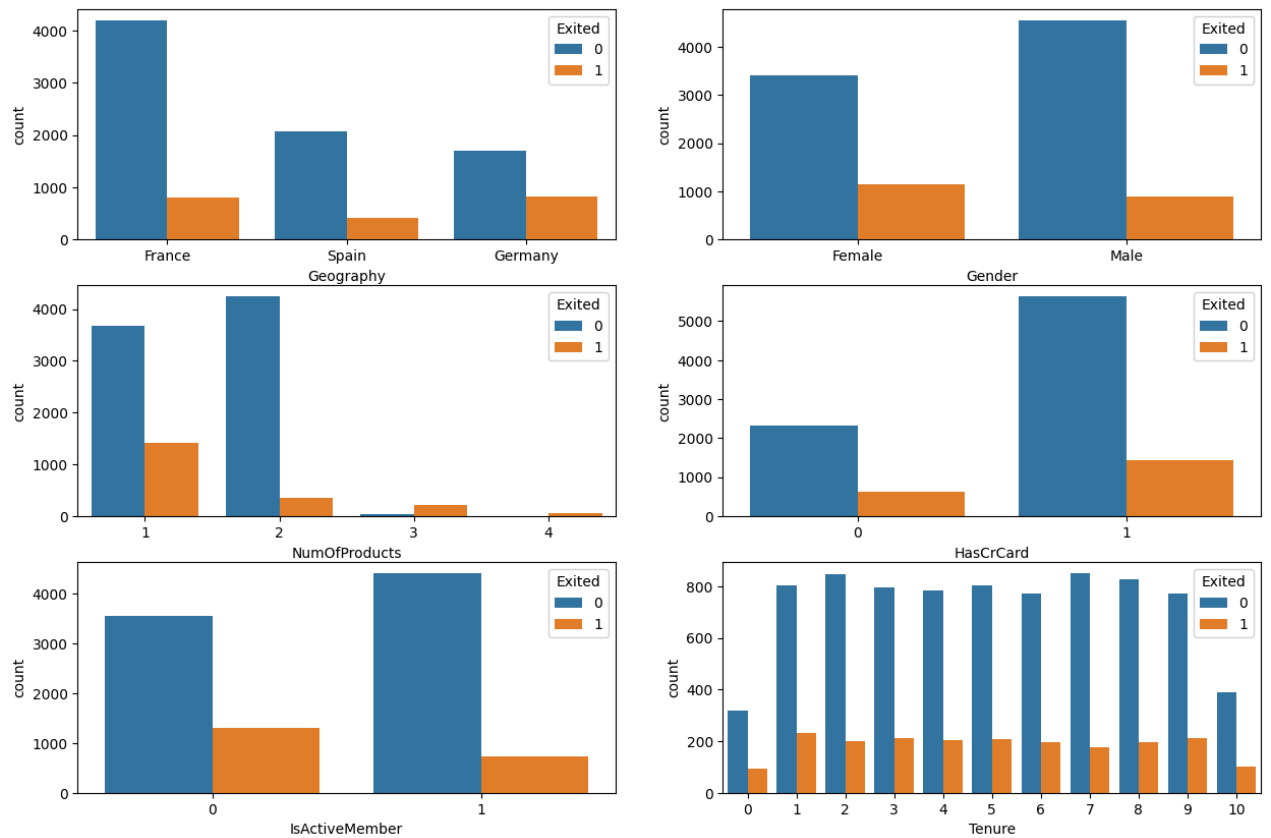
```
sns.countplot(x = 'Geography', hue= "Exited" , data = df)
plt.show()
```



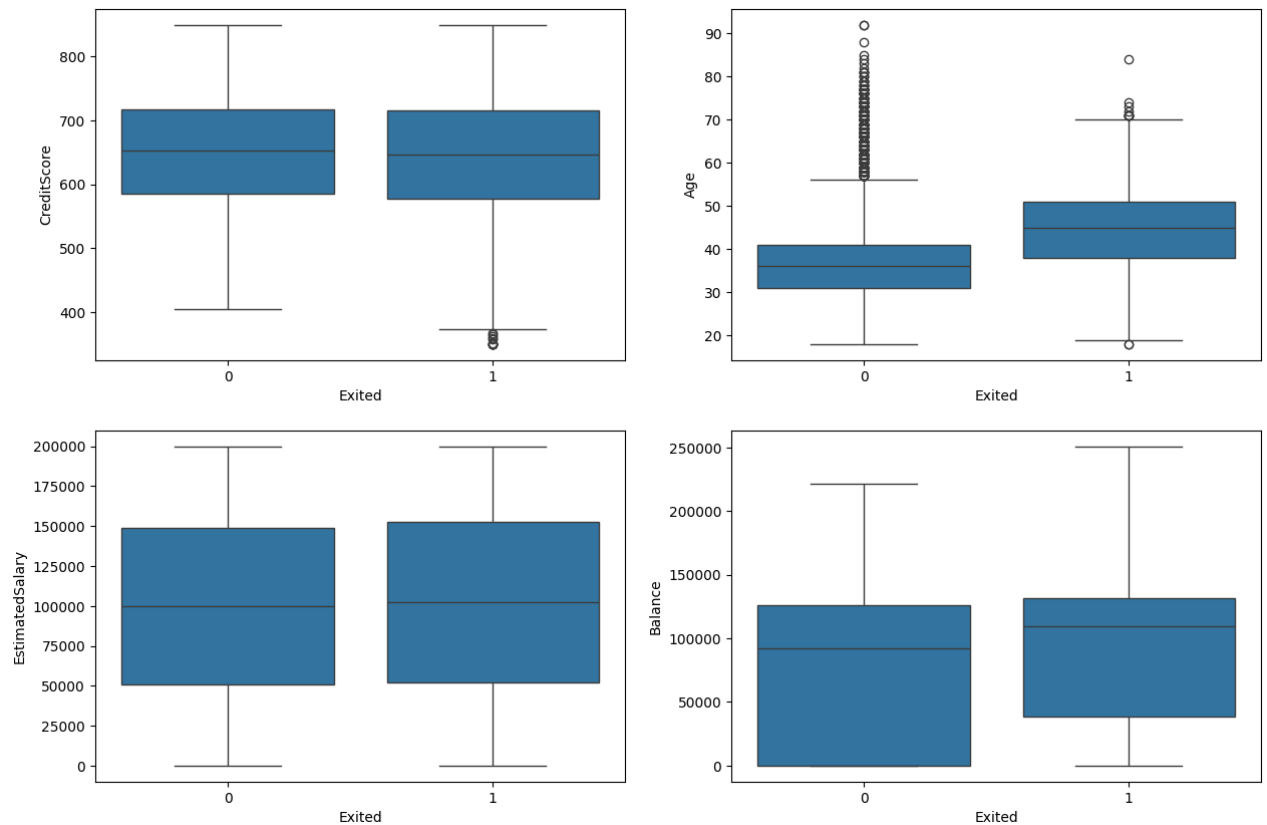
```
df.head()
```

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

```
#subplot(3x2)
fig, ax = plt.subplots(3,2, figsize = (15,10))
sns.countplot(x = "Geography", hue= "Exited", data = df, ax= ax[0,0])
sns.countplot(x = "Gender", hue= "Exited", data = df, ax= ax[0,1])
sns.countplot(x = "NumOfProducts", hue= "Exited", data = df, ax= ax[1,0])
sns.countplot(x = "HasCrCard", hue= "Exited", data = df, ax= ax[1,1])
sns.countplot(x = "IsActiveMember", hue= "Exited", data = df, ax= ax[2,0])
sns.countplot(x = "Tenure", hue= "Exited", data = df, ax= ax[2,1])
plt.show()
```



```
figure, ax = plt.subplots(2,2, figsize= (15,10))
sns.boxplot(x="Exited", y="CreditScore", data =df, ax =ax[0,0])
sns.boxplot(x="Exited", y="Age", data =df, ax =ax[0,1])
sns.boxplot(x="Exited", y="EstimatedSalary", data =df, ax =ax[1,0])
sns.boxplot(x="Exited", y="Balance", data =df, ax =ax[1,1])
plt.show()
plt.tight_layout()
```



<Figure size 640x480 with 0 Axes>

```
df.columns
```

```
Index(['CreditScore', 'Geography', 'Gender', 'Age', 'Tenure', 'Balance',
      'NumOfProducts', 'HasCrCard', 'IsActiveMember', 'EstimatedSalary',
      'Exited'],
      dtype='object')
```

```
df = pd.get_dummies(df, columns= ["Geography", "Gender"], drop_first= True)
```

```
-----
KeyError                                Traceback (most recent call last)
/tmp/ipykernel_3425/1451776339.py in <cell line: 0>()
----> 1 df = pd.get_dummies(df, columns= ["Geography", "Gender"], drop_first= True)

~
~
~
3 frames
/usr/local/lib/python3.12/dist-packages/pandas/core/indexes/base.py in _raise_if_missing(self, key, indexer, axis_name)
    6247         if nmissing:
    6248             if nmissing == len(indexer):
-> 6249                 raise KeyError(f"None of [{key}] are in the [{axis_name}]")
    6250
    6251         not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique())
KeyError: "None of [Index(['Geography', 'Gender'], dtype='object')] are in the [columns]"
```

```
df.head()
```

CreditScore Age Tenure Balance NumOfProducts HasCrCard IsActiveMember EstimatedSalary Exited Geography_Germany

```
plt.figure(figsize= (15,10))
sns.heatmap(df.corr(), annot= True, cmap= "coolwarm")
plt.show()
```

